

LECTURE 05 · REVIEW NOTES

Long/Short and Optimal Allocation

What changes when portfolio weights go below zero or above 100%, and how to choose the optimal combination of two risky assets.

About these notes

These notes review the material of Lecture 05. Each concept is explained in full, and each numerical example is worked out step by step, with every intermediate figure shown. The two graded activities completed during class are summarized in general terms and marked as in-class activities.

1. A real long/short fund: the AQR case

Until now every portfolio weight has been between 0% and 100%. Many real funds do not work that way. The AQR Long–Short Equity Fund is a systematic strategy that **buys** stocks expected to outperform and **sells short** stocks expected to underperform. As of September 21, 2026 it managed \$9.521 billion, and on June 30, 2026 it held 933 long positions and 960 short positions. The fund often holds long and short positions in the same sector, because its two sources of return, **stock selection** and **market exposure**, are separate decisions: it can bet on which bank will beat another bank without betting on the direction of the banking sector as a whole.

Two measures describe such a portfolio. **Gross exposure** adds the absolute size of all positions and measures how much capital is at work. **Net exposure** subtracts the shorts from the longs and measures the directional bet on the market.

WORKED EXAMPLE 1

Gross and net exposure per \$100 of fund assets

On June 30, 2026, for every \$100 of net assets the fund held \$232.41 of long stock positions, \$223.81 of short stock positions and \$43.07 of equity futures.

Step	Calculation	Result
1	Gross stock exposure = long + short = 232.41 + 223.81	456.22% of net assets
2	As a multiple of net assets: $456.22 \div 100$	4.56×
3	Net stock exposure = long – short = 232.41 – 223.81	8.60%
4	Futures add a further +43.07% of exposure	Not included in 1-3

The fund's stock positions are large, but they nearly offset each other, leaving a much smaller directional balance. Two cautions apply. The stock-only net figure is not the fund's total exposure or its beta, since the futures add exposure of their own; and gross exposure

is not a debt ratio. More importantly, **low net exposure does not mean low risk**. Long and short positions can move against the fund at the same time, for example if the stocks it owns fall while the stocks it is short rise. Gross positions, correlations and financing costs all still matter.

AQR fund data are from the fund's published materials as presented in Lecture 05; an educational case, not a recommendation.

2. A simple model: the 120/20 portfolio

The mechanics can be seen with just two assets. An investor starts with \$100 of equity, sells short \$20 of asset B, and uses the \$100 plus the \$20 of proceeds to buy \$120 of asset A. The short sale generates cash, but also an obligation: asset B must be bought back later. If B rises, the short position loses value, so short-sale proceeds are not free wealth. (The model ignores collateral restrictions and trading costs.)

Position	Dollars	Weight
Long asset A	\$120	$w_A = 120 \div 100 = 1.20$
Short asset B	-\$20	$w_B = -20 \div 100 = -0.20$
Sum (equity)	\$100	$w_A + w_B = 1.00$
Net exposure	$120 - 20$	100%
Gross exposure	$120 + 20$	140%

The weights still sum to one, but individual positions can now exceed the investor's equity. The same portfolio formulas apply unchanged; what changes is how the signs work out.

WORKED EXAMPLE 2

Expected return of the 120/20 portfolio

Asset A has $E[r] = 10\%$ and $\sigma = 20\%$; asset B has $E[r] = 5\%$ and $\sigma = 15\%$; their correlation is $\rho = 0.30$.

Step	Calculation	Result
1	$E[r_p] = w_A E[r_A] + w_B E[r_B] = 1.20 \times 10\% + (-0.20) \times 5\%$	12% - 1%
2	Expected return of the 120/20 portfolio	11.00%
3	Gain relative to holding A alone: $\Delta E[r] = 0.20 \times (E[r_A] - E[r_B]) = 0.20 \times (10\% - 5\%)$	+1.00 pp

The extra return comes entirely from the **spread** between the two assets: the investor earns A's return on the extra 20% and pays B's return on the short. If the ranking were reversed and B were expected to outperform A, the same trade would lower the expected return.

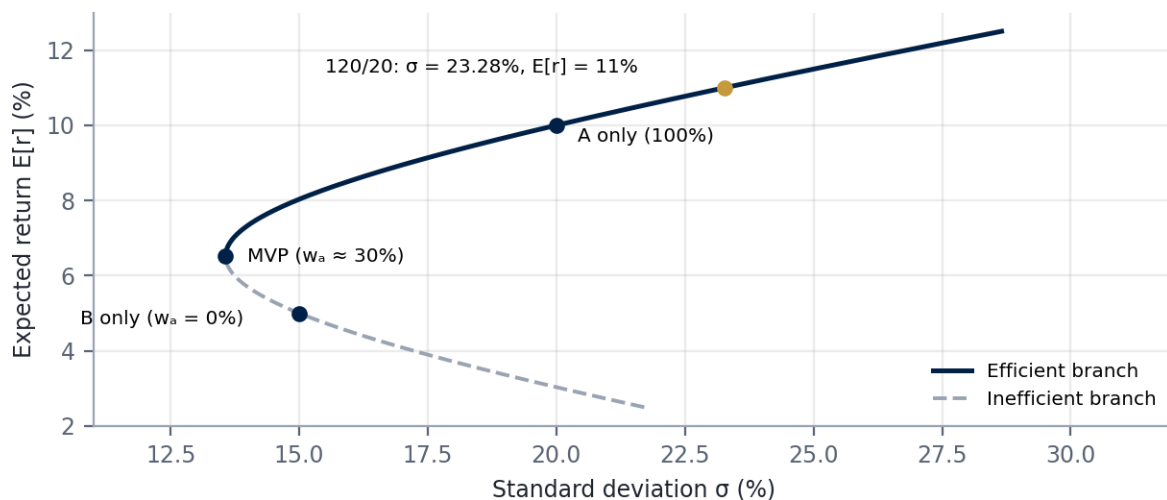
WORKED EXAMPLE 3

Volatility of the 120/20 portfolio

Risk is not a weighted average. Working in percentage units throughout:

Step	Calculation	Result
1	A term: $w_A^2 \sigma_A^2 = 1.20^2 \times 20^2 = 1.44 \times 400$	576.0
2	B term: $w_B^2 \sigma_B^2 = (-0.20)^2 \times 15^2 = 0.04 \times 225$	9.0
3	Cross-term: $2w_A w_B \rho \sigma_A \sigma_B = 2 \times 1.20 \times (-0.20) \times 0.30 \times 20 \times 15$	-43.2
4	Variance = $576.0 + 9.0 - 43.2$	541.8 (%) ²
5	$\sigma_p = \sqrt{541.8}$	23.28%

Both squared-weight terms are positive whatever the signs of the weights. The cross-term is negative here because one weight is negative while the correlation is positive: the short position in B partly hedges the long position in A. Even so, volatility rises from 20% for A alone to 23.28%, because the portfolio holds 120% in the riskier asset. Leverage does not change risk by the same amount for every pair of assets; the outcome depends on the volatilities and the correlation. Real portfolios also incur short-borrow fees, dividends owed on the short position and trading costs.



The opportunity set for A and B with weights on A from -50% to 150%. Allowing short positions extends the curve beyond both assets. Only the upper branch, above the minimum-variance portfolio, is efficient.

Once short positions are allowed, the opportunity set no longer stops at the two assets. Weights on A above 100% (shorting B) extend the efficient branch beyond A toward higher return and higher risk; weights below 0% (shorting A) extend the lower, inefficient branch beyond B.

IN-CLASS ACTIVITY · Activity 1: a leveraged portfolio

Each student received individual long/short weights for the two assets and computed the portfolio's expected return and volatility, applying the same formulas as Examples 2 and 3 to a different case.

3. Choosing the optimal risky portfolio

With many possible portfolios on the frontier, the question becomes which one to hold. When a risk-free asset is available, every risky portfolio P defines a capital allocation line from r_f through P, whose slope is P's Sharpe ratio:

$$\text{Sharpe} = (E[r_p] - r_f) \div \sigma_p$$

The best choice is the portfolio with the **steepest attainable line**, the **tangency portfolio**. It offers the highest expected excess return per unit of volatility, which makes it the optimal *risky* mix. How much of the investor's money to place in it, versus the risk-free asset, is a separate decision that depends on the investor's risk tolerance. "Optimal" here means the maximum Sharpe ratio under the model's assumptions.

Setting up the problem

With two risky assets there is a single choice variable, w_A , since $w_B = 1 - w_A$. The problem is to choose w_A to maximize:

$$[w_A E[r_A] + (1 - w_A) E[r_B] - r_f] \div \sqrt{w_A^2 \sigma_A^2 + (1 - w_A)^2 \sigma_B^2 + 2w_A(1 - w_A)\rho\sigma_A\sigma_B}$$

The inputs are two expected returns, two volatilities, the correlation (or the covariance $\rho\sigma_A\sigma_B$) and the risk-free rate. They are inputs, not decisions: the problem chooses weights, it does not forecast returns. The model allows short positions and has no position limits or trading costs.

The closed-form solution

Writing $R_A = E[r_A] - r_f$ and $R_B = E[r_B] - r_f$ for the excess returns and $c = \rho\sigma_A\sigma_B$ for the covariance, the Sharpe-maximizing weights are:

$$w_A^* = (R_A\sigma_B^2 - R_Bc) \div [R_A\sigma_B^2 + R_B\sigma_A^2 - (R_A + R_B)c], \quad w_B^* = 1 - w_A^*$$

WORKED EXAMPLE 4**The tangency portfolio of XLK and XLV**

Illustrative inputs (not current estimates): XLK (A) has $E[r] = 12\%$ and $\sigma = 20\%$; XLV (B) has $E[r] = 8\%$ and $\sigma = 15\%$; $\rho = 0.30$ and $r_f = 3\%$. All figures are in percent.

Step	Calculation	Result
1	Excess returns: $R_A = 12 - 3$; $R_B = 8 - 3$	$R_A = 9$, $R_B = 5$
2	Covariance $c = \rho\sigma_A\sigma_B = 0.30 \times 20 \times 15$	$c = 90$
3	Numerator = $R_A\sigma_B^2 - R_Bc = 9 \times 225 - 5 \times 90$ $= 2,025 - 450$	1,575
4	Denominator = $R_A\sigma_B^2 + R_B\sigma_A^2 - (R_A + R_B)c =$ $2,025 + 2,000 - 14 \times 90$	2,765
5	$w_A^* = 1,575 \div 2,765$; $w_B^* = 1 - w_A^*$	56.96% XLK, 43.04% XLV
6	$E[r] = 0.5696 \times 12\% + 0.4304 \times 8\% = 6.84\%$ $+ 3.44\%$	10.28%
7	$\sigma^2 = 0.5696^2 \times 400 + 0.4304^2 \times 225 + 2 \times$ $0.5696 \times 0.4304 \times 90 = 129.8 + 41.7 + 44.1$ $= 215.6$	$\sigma = 14.68\%$
8	Sharpe = $(10.28 - 3) \div 14.68$	≈ 0.496

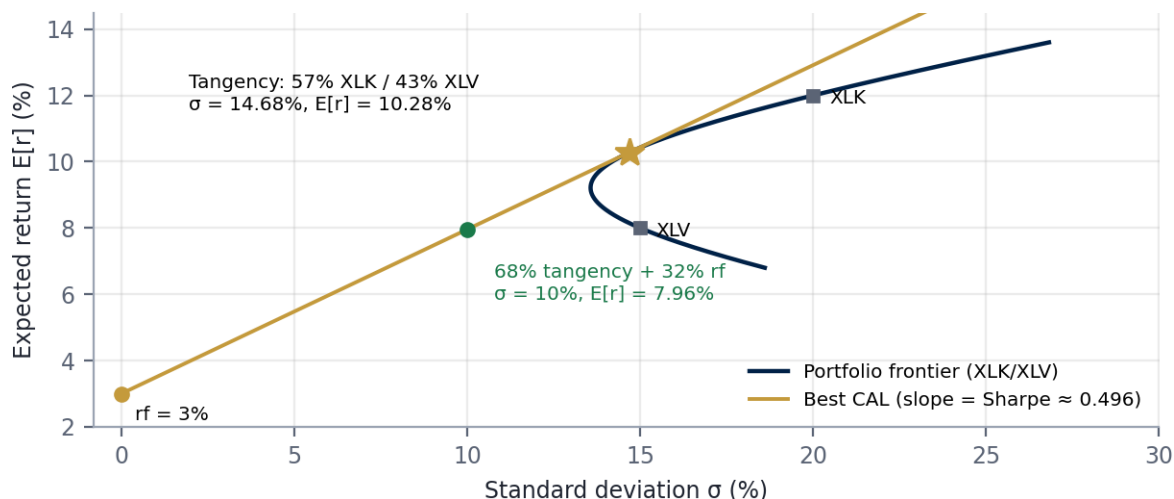
For comparison, XLK alone has a Sharpe ratio of $(12 - 3) \div 20 = 0.45$ and XLV alone $(8 - 3) \div 15 = 0.33$. The mix beats both because the correlation of 0.30 lowers risk more than it lowers return. In this case both optimal weights are positive, so no short sale is needed.

WORKED EXAMPLE 5**From the optimal risky portfolio to the investor's portfolio**

Suppose an investor wants a total volatility of 10%. The investor holds the tangency portfolio and the risk-free asset in the proportions that hit that target. (Illustrative.)

Step	Calculation	Result
1	Fraction in the tangency portfolio: $y = 10\% \div 14.68\%$	$y = 68.1\%$
2	Fraction in the risk-free asset = $1 - 0.681$	31.9%
3	$E[r] = 3\% + 0.681 \times (10.28\% - 3\%)$	7.96%
4	Sharpe of the combination = $(7.96 - 3) \div 10$	0.496, unchanged

Every combination along the line keeps the tangency portfolio's Sharpe ratio. A more aggressive investor would choose y above 68%, even above 100% by borrowing at the risk-free rate, but the risky mix itself would not change. This is why the choice of the risky portfolio and the choice of how much risk to take are two separate decisions.



The XLK/XLV frontier, the tangency portfolio and the best capital allocation line from $r_f = 3\%$. The green point is the 10%-volatility combination from Example 5.

IN-CLASS ACTIVITY · Activity 2: find the optimal portfolio

Using the same pair of assets as in Activity 1, each student found the risky allocation with the highest Sharpe ratio by applying the tangency formula to an individual set of inputs.

4. What if short selling is not allowed?

The closed-form solution assumes that weights can take any value. If the unconstrained solution calls for a short position, for example w_B^* below zero, and short selling is prohibited, that answer is no longer feasible. The feasible set shrinks, and the constrained problem must be solved instead, typically with a numerical optimizer, as in the long-only frontier of Lecture 04. With two assets the Sharpe ratio rises steadily toward the unconstrained optimum, so the constrained optimum sits at the nearest boundary: 100% in the asset the unconstrained solution was buying with the short-sale proceeds. When the unconstrained weights are already between 0% and 100%, as in Example 4, the constraint does not bind and the answer is unchanged.

5. Key takeaways

Allowing long/short positions. Weights can be negative or exceed 100% while still summing to one. Gross exposure measures how much capital is at work; net exposure measures the directional bet. Low net exposure does not guarantee low risk.

Measuring return and risk. Expected return is linear in the weights, so a short position adds the spread between the two assets' returns. Volatility depends on the squared weights and the covariance, and a negative weight flips the sign of the cross-term.

Choosing the risky mix. The maximum Sharpe ratio identifies the tangency portfolio, which is the same for every investor using the same inputs. Risk preference then determines how that portfolio is combined with the risk-free asset.